



*The University of Azad Jammu and Kashmir
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COURSE TITLE	Computer Architecture and Logic Design
COURSE CODE	CS-1205
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LAB TITLE	Combinational Circuit(Full Adder,Half Subtractor,Full Subtractor)

Department of Software Engineering

Full Adder

Definition

A **Full Adder** is a digital circuit that performs the addition of three binary inputs:

- **A**: First bit
- **B**: Second bit
- **Cin** (Carry-in): Carry from the previous stage

It produces two outputs:

- **Sum**: Result of the addition
- **Cout** (Carry-out): Carry generated for the next stage

Truth Table

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Logic Expressions

- **Sum** = $A \oplus B \oplus \text{Cin}$
- **Cout** = $(A \cdot B) + (B \cdot \text{Cin}) + (A \cdot \text{Cin})$

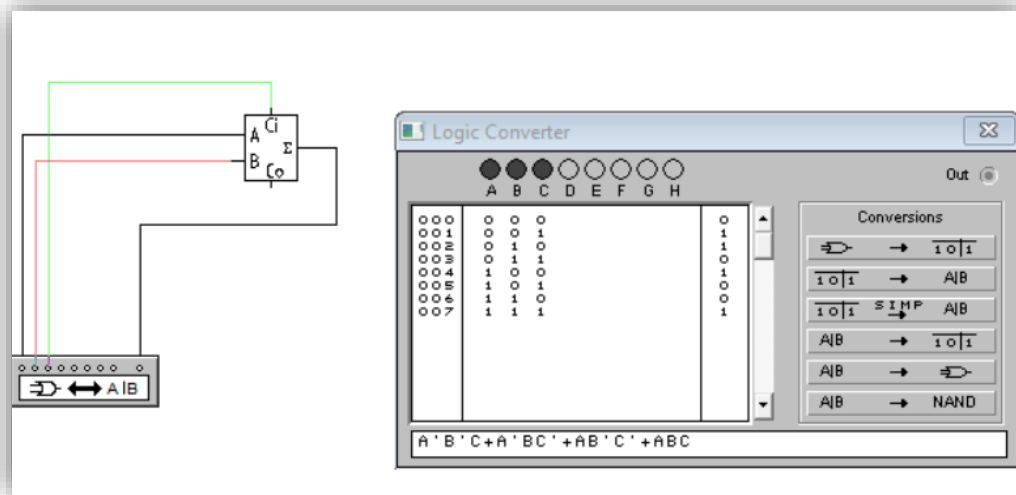
Implementation

- Can be built using two **Half Adders** and one **OR** gate.

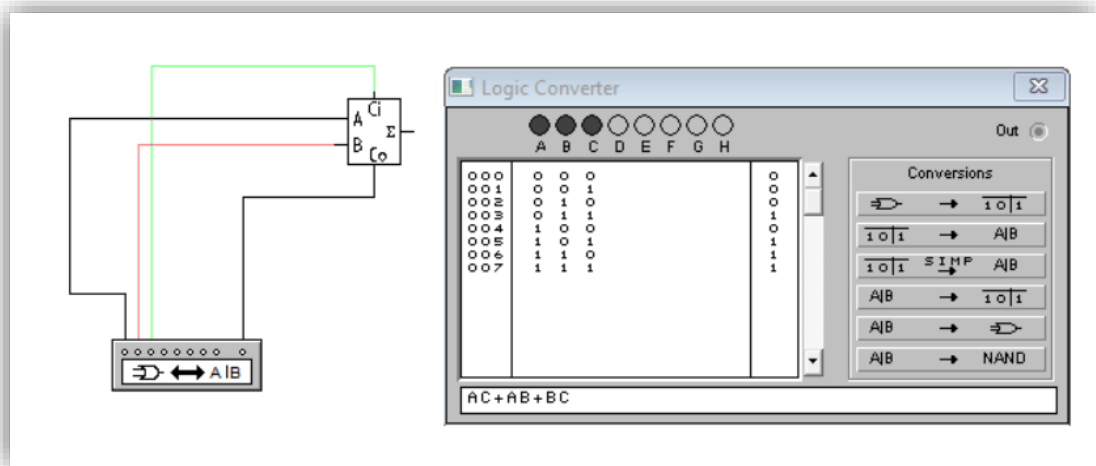
- Widely used in multi-bit binary addition by connecting several full adders in series (Ripple Carry Adder).

Applications

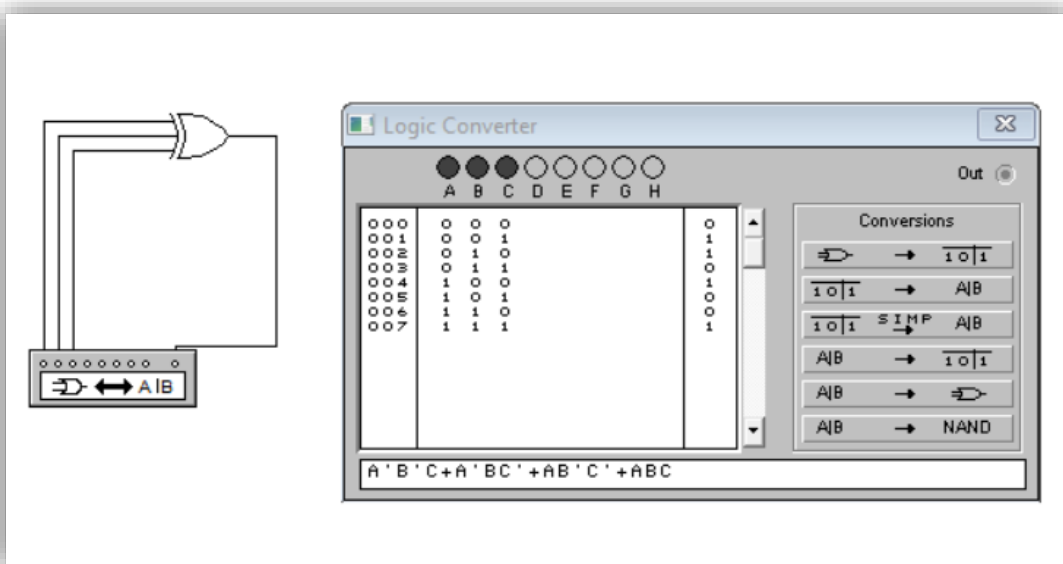
- Arithmetic Logic Units (ALUs)
- Digital calculators
- Processors and microcontrollers
- Multipliers and digital signal processing



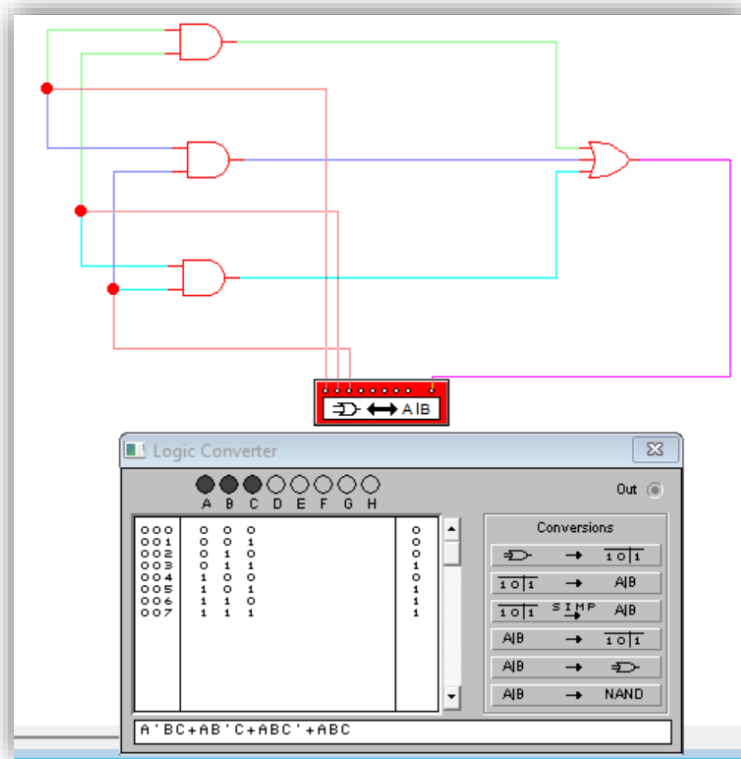
For Sum truth table, equation and logic Converter.



For Carry truth table, equation and logic converter.



Circuit Diagram of Full Adder



Circuit Diagram

of Carry

Half Subtractor

Definition

A **Half Subtractor** is a **combinational logic circuit** that performs subtraction between two binary digits.

It has **two inputs**:

- **A**: Minuend (the number from which another number is subtracted)
- **B**: Subtrahend (the number to be subtracted)

It produces **two outputs**:

- **Difference (D)**: The result of subtraction ($A - B$)
- **Borrow (Bo)**: Indicates if a '1' has been borrowed from the next higher bit
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Truth Table

A	B	Difference (D)	Borrow (Bo)
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

Logic Expressions

- **Difference** = $A \oplus B$
- **Borrow** = $\neg A \cdot B$

Where:

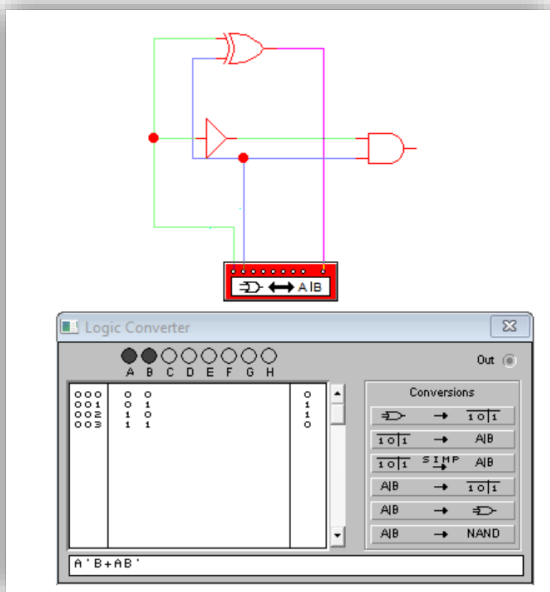
- $\oplus \rightarrow$ XOR operation
- $\neg A \rightarrow$ NOT A
- $\cdot \rightarrow$ AND operation

Implementation

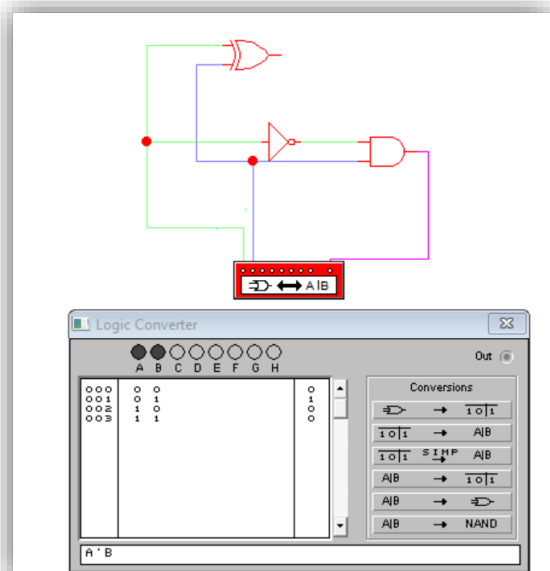
- The **Difference** output is implemented using an **XOR gate**.
- The **Borrow** output is implemented using a **NOT gate** and an **AND gate**.

Applications

- Binary subtraction in small-scale circuits
- Used in building **Full Subtractors**
- Arithmetic operations in digital systems



Truth table and Equation for Difference



Truth table and Equation for Barrow

Full Subtractor

Definition

A **Full Subtractor** is a **combinational logic circuit** that performs subtraction of three binary bits:

- **A**: Minuend (the main number)
- **B**: Subtrahend (the number to be subtracted)
- **Bin**: Borrow in (borrow from the previous stage)

It produces two outputs:

- **Difference (D)** : Result of subtraction ($A - B - \text{Bin}$)
- **Bout** (Borrow out): Borrow generated for the next stage

Truth Table

A	B	Bin	Difference (D)	Bout
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

Logic Expressions

- **Difference** = $A \oplus B \oplus \text{Bin}$
- **Bout** = $(\neg A \cdot B) + (B \cdot \text{Bin}) + (\neg A \cdot \text{Bin})$

Where:

- $\oplus \rightarrow$ XOR operation
- $\neg A \rightarrow$ NOT A
- $\cdot \rightarrow$ AND operation
- $+$ $\rightarrow$ OR operation

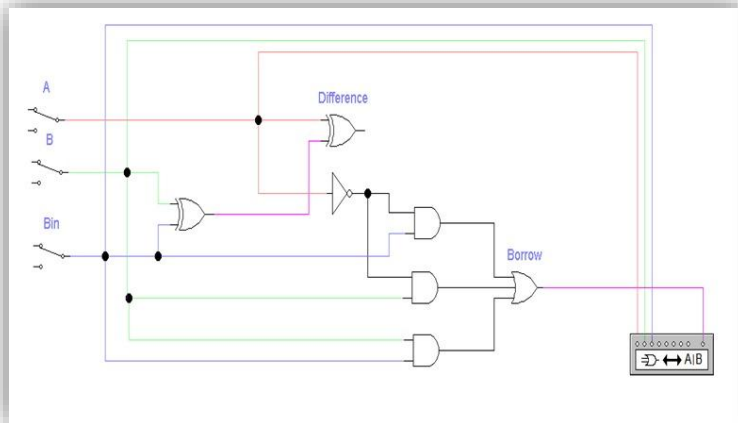
Implementation

- Built using **two Half Subtractors** and an **OR gate**.
- The first Half Subtractor finds the initial difference and borrow.
- The second Half Subtractor processes the borrow-in, and the OR gate combines borrows to produce the final borrow-out.

Applications

- Multi-bit binary subtraction
- Used in **Arithmetic Logic Units (ALUs)**
- Digital processors for subtraction operations

Circuit Diagram and Equation -Borrow



Logic Converter

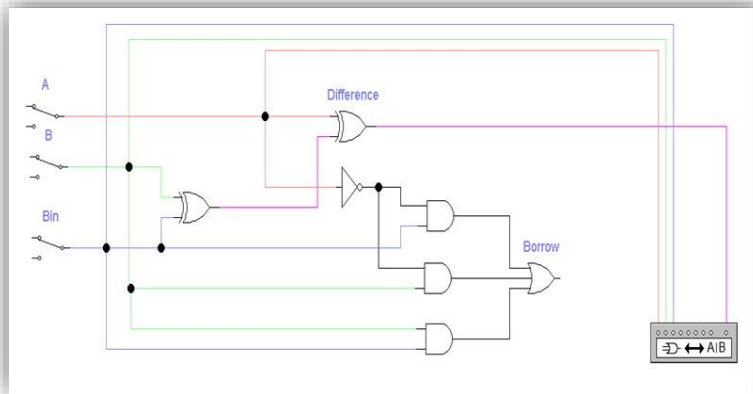
	A	B	C	D	E	F	G	H
000	0	0	0					0
001	0	0	1					1
002	0	1	0					1
003	0	1	1					1
004	1	0	0					1
005	1	0	1					0
006	1	1	0					1
007	1	1	1					1

Conversions

Out

$A'B'C + A'BC' + AB'C' + ABC$

Circuit Diagram and Equation – Difference



Logic Converter

	A	B	C	D	E	F	G	H
000	0	0	0					0
001	0	0	1					1
002	0	1	0					1
003	0	1	1					1
004	1	0	0					1
005	1	0	1					0
006	1	1	0					1
007	1	1	1					1

Conversions

Out

$A'B + A'C + BC$

Best Of Luck